

Malcolm Greaves

MACHINE LEARNING SYSTEMS ENGINEER

U.S. Citizen, San Francisco Bay Area

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Experience

NVIDIA

New York City

SENIOR SYSTEMS ENGINEER IV

October 2023 - Present

- Created first automated, API-driven hardware and software stack testing service in NVIDIA Enterprise for BioNeMo. Ensuring that BioNeMo is compatible with dozens of NVIDIA GPUs, processor types, operating systems, different versions of PyTorch and CUDA.
- Ongoing mentoring and skills-teaching to two dozen engineers and applied scientists. Weekly group demos and 1:1 teaching sessions.
- Built-out BioNeMo's internal and external CI services to support open source development and internal testing and profiling. Led coordination efforts with internal infrastructure teams, optimizing scheduling and builds in CI, reducing build times by 30%.

Scale AI, Inc.

San Francisco & New York City

STAFF ML ENGINEER

March 2020 - October 2023

- Created Scale's first document processing models and supporting infra., securing a 7-figure contract & founding Scale Document.
- Creator, GPU utilization working group: up leveled other ML eng to optimize their team's service from 20% to 86% GPU utilization.
- Advised and mentored ML engineer who optimized Kubernetes service start time from 2 minutes to under 30 seconds.
- Created ML model serving framework used by all ML engineers and hundreds of internal and external services.
- Optimized internal LLM services, achieving 10x speedup in end-to-end inference.
- Re-wrote real-time image generation + RAG pipeline with 10M+ requests/day with efficient vector database, reducing costs by 50%.
- Optimized batch embedding generation service used by multiple internal teams; decreased cost by 5x without increasing run time.

Change Healthcare

Emeryville, CA

SENIOR DATA SCIENTIST

November 2018 - February, 2020

- Developed deep NN model for predicting missing charges & deployed to US's largest medical insurance payment network.
- Increased scientist productivity by 25x with internal model training & serving framework.

Volley Labs, Inc.

San Francisco, CA

SENIOR RESEARCH ENGINEER, ML TECH LEAD

December 2016 - October 2018

- End-to-end ownership & development of multiple-choice question generation system from unsupervised to supervised neural network approach. Product-differentiating feature that helped Volley stand-out and make its first sale to JP Morgan Chase.
- Technical lead of engineering and ML team. Data pipelines in Airflow; deployed ML models using Keras, Spacy, Tensorflow; Python 3.

Nitro Software, Inc.

San Francisco, CA

RESEARCH ENGINEER

March 2015 - October 2016

- Created novel machine learning based solution for automatic form field detection (FFD) and semantic classification. End-to-end service development & deployment to Nitro Cloud.

Alpine Data Labs

San Francisco, CA

SOFTWARE ENGINEER, MACHINE LEARNING

Jun. 2014 - Mar. 2015

- Algorithm optimization for large-scale data processing & modeling in Spark: L-BFGS, random forest models, feature encoding, etc.

Education

Carnegie Mellon University

Pittsburgh, PA USA

B.Sc. AND M.Sc. IN COMPUTER SCIENCE

Aug. 2009 - May 2014

- Graduated with School of Computer Science Honors. GPA 3.5 (undergrad and graduate).
- Master's Thesis in semantic relation extraction from unstructured text with probabilistic logic & SVMs: <http://goo.gl/DzMr6c>

Work Portfolio

Languages	Proficient: Python, Rust, Scala, Go, SQL, Java, BASH, C; Familiar: Typescript, LaTeX, OCaml, C++11, CUDA
Machine Learning & Data	PyTorch, Pandas, HuggingFace, TensorFlow, OpenCV, LangChain, numpy, scipy, scikit-learn, matplotlib, Onnx, burn, NVIDIA Triton, Deepspeed, Spark, DynamoDB, Postgres, MongoDB, Snowflake, Elasticache,
Infrastructure & Backend	Kubernetes, Datadog, Terraform, Istio, Temporal, Celery, Redis, Kafka, RabbitMQ, qdrant, milvus, tokio, rayon, serde, axum, crossbeam, datafusion, Protobuf, gRPC, OpenAPI, Flask, FastAPI, MinIO,